Kyeongseon (Ella) Kim

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Research Interests

My research explores how visual structure, interaction, and cultural context contribute to multimodal understanding. I uncover informative signals, such as semantic structure in vector graphics and implicit cues in user behavior, and design representations that help models and agents understand meaning and users' needs. My goal is to make unfamiliar digital environments more approachable, helping people of all ages, abilities, and backgrounds explore and adapt with confidence.

Education

KAIST — Daejeon, Korea
Ph.D. Student, School of Computing (Advisor: Prof. Tae-Hyun Oh) — Sep. 2024 – Present

Chung-Ang University — Seoul, Korea
M.S. in Computer Science (Advisor: Prof. Joongheon Kim, now at Korea University) — Sep. 2017 – Aug. 2019
B.S. in Computer Science and Engineering — Mar. 2013 – Feb. 2017

Research & Industry Experience

POSTECH — Pohang, Korea
AI Research Scientist (Lab of Prof. Tae-Hyun Oh) — 2023 – 2024

Upstage (company behind the Solar LLM) — Seoul, Korea
AI Contents & Education Manager / AI Pack User Guide Project Manager — 2022

LG Electronics (CTO Office AI Lab) — Seoul, Korea
AI Research Engineer (Team led by Prof. Joseph Lim, USC) — 2020 – 2022

- Research topic: computer vision, scene graph generation
- Collaborated with LG Toronto AI Lab on floor plan generation for object placement
- Designed a driver action recognition system (Hyundai, Daimler)

EASYWUM — Seoul, Korea
Tech Lead, Smart IoT Indoor Garden Startup — 2016 – 2017

Awards & Scholarships

Korean Government-Funded Scholarship (Ph.D. Program), KAIST — 2025 – Present

Best Student Presentation Award, IPIU (for mEOL [2], WACV 2026) — 2025

Selected Research & Publications

"" denotes equal contribution*

[1] **IGG: A Benchmark for Interactive GUI Grounding under Visibility Constraints.**
K. S. Kim*, J. Son*, and T.-H. Oh.
Conference on Empirical Methods in Natural Language Processing (**EMNLP 2026, Main**)
Introduced a benchmark for evaluating whether GUI agents can reveal and ground targets through interaction under visibility constraints.
Also presented at ICML 2026 Workshop on Compositional Learning: Safety, Interpretability, and Agents

[2] **mEOL: Training-Free Instruction-Guided Multimodal Embedder for Vector Graphics and Image Retrieval.**
K. S. Kim, S.-E. Baek, J.-M. Lee, and T.-H. Oh.
IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV 2026**)
Developed training-free multimodal embeddings that capture visual structure and implicit semantic associations across text, images, and SVG code for concept-based icon retrieval.
Accepted as a Round 1 paper (6.4% acceptance rate); Best Student Presentation Award, IPIU 2025
Also presented at ICCV 2025 Workshop on Graphic Design Understanding and Generation (GDUG)

[3] **CuPS: Measuring Cultural Preference Signatures in LLM/VLM Agents and Their Steering by Profile Memories.**
K. S. Kim, G. Kim, J. Chang, H. Kim, and T.-H. Oh.
Workshop on Culture × AI: Evaluating AI as a Cultural Technology (**ICML 2026 Workshop**)
Benchmarked model differences in symbol interpretation and concept association, and examined how cultural cues in saved user context shape agents' responses.

[4] **Understanding User Difficulties in Digital Interfaces**
Work in progress
Investigating how multimodal models can identify users' interface difficulties from screens, actions, and user utterances, toward personalized assistance that supports older adults' self-efficacy and autonomy.

[5] **SVHalluc: Benchmarking Speech–Vision Hallucination in Audio-Visual Large Language Models.**
C. Zhang, **K. S. Kim**, C. Liu, and T.-H. Oh.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2026**)

[6] **Zero-shot Depth Completion via Test-time Alignment with Affine-invariant Depth Prior.**
H. Lee, **K. S. Kim**, B.-K. Kwon, and T.-H. Oh.
AAAI Conference on Artificial Intelligence (**AAAI 2025**)
Qualcomm Innovation Fellowship 2025 winning paper

[7] **Personalized Online Live Video Streaming Using Softmax-Based Multinomial Classification.**
K. S. Kim, D. Kwon, J. Kim, and A. Mohaisen.
Vol. 9, No. 11, Article 2297 (**Applied Sciences 2019**)

[8] **Self-Adaptive Machine Learning Operating Systems for Security Applications.**
K. S. Kim, D. Kwon, Y. Kim, J. Kim, and J. Kim.
IEEE International Conference on Acoustics, Speech and Signal Processing (**IEEE ICASSP 2018**)

Invited Talks & Workshops

AI Agents for Everyday Work: A Hands-on Workshop for Non-developers, Samsung Multicampus	2025
AI Talk for High School Sophomores and Juniors, Jeonbuk Science High School	2023
MODUCON: AI Community Organizing, Yonsei University / Lab for Everyone	2022
TensorFlow Everywhere Korea Keynote: What's New in TF 2.4, Google	2021
AI Talk for High School Sophomores and Juniors, Hanseong Science High School (Seoul, South Korea)	2021

Selected Leadership & Teaching

AGI Korea (formerly TensorFlow Korea) – Founder & Organizer (66K+ members)	2020 – Present
Google TensorFlow User Group (TFUG) – Organizer (Global)	2020 – Present
TensorFlow Everywhere – General Organizer (Global)	Feb. 2021
Software Maestro (Ministry of Science and ICT, Korea) – Mentor	2022 – 2024
Upstage × Naver Boostcamp AI Tech – Organizer & Mentor	2021 – 2022
Lecturer, Deep Learning Project Management Process (Naver Boostcamp AI Tech)	2022
SW Health Contents Development, College of Nursing, Chung-Ang University – Lecturer	2018

Selected Patents

KR 10-1894314 (Granted): Method and Apparatus for Guiding a Recommended Parking Location
(tech. transferred, Nov. 2024)

KR 10-2019-0005307: Method and System for Video Streaming Service